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Subject : DSA

Unit 3, S8, SLO1 - Queue Introduction

1. A queue follows the ____ principle.

Answer: FIFO (First In, First Out)

2. In a queue, insertion is performed at the ____ end.

Answer: rear

3. In a queue, deletion is performed at the ____ end.

Answer: front

4. A ____ queue is a linear data structure with a fixed size.

Answer: linear (array-based)

5. A ____ queue overcomes the limitation of fixed size by wrapping around.

Answer: circular

6. A ____ queue allows insertion and deletion from both ends.

Answer: deque (double-ended)

7. ____ queue gives priority to elements based on certain criteria.

Answer: Priority

8. Queues are used in ____ scheduling in operating systems.

Answer: CPU / process / I/O

9. In a queue, if the rear is just behind the front in a circular manner, the queue is said to be ____.

Answer: full

10. A ____ is used to implement queues in programming languages like C.

Answer: array or linked list

11. The queue operation that checks whether it is full is called ____.

Answer: isFull()

12. The queue operation that checks whether it is empty is called ____.

Answer: isEmpty()

13. In breadth-first search (BFS) of a graph, ____ is used to keep track of the next node to visit.

Answer: a queue

14. In printers, tasks are managed using a ____ to print in the order they arrive.

Answer: print spooler / queue

15. A queue where new elements are added based on priority rather than position is called a ____.

Answer: priority queue